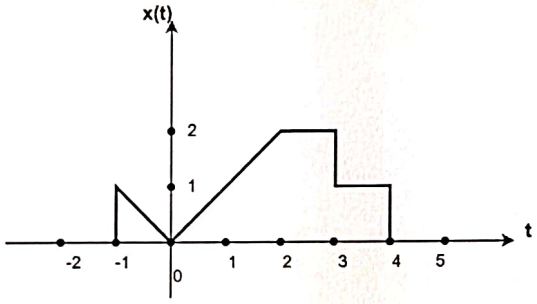
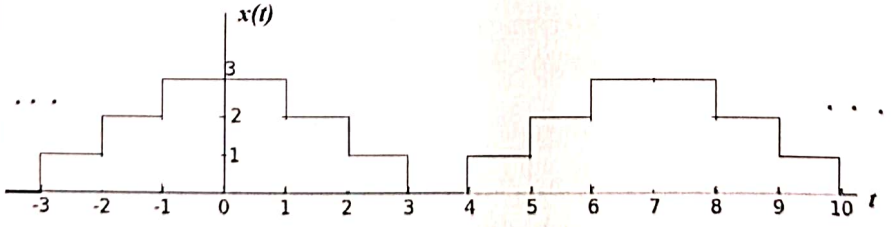
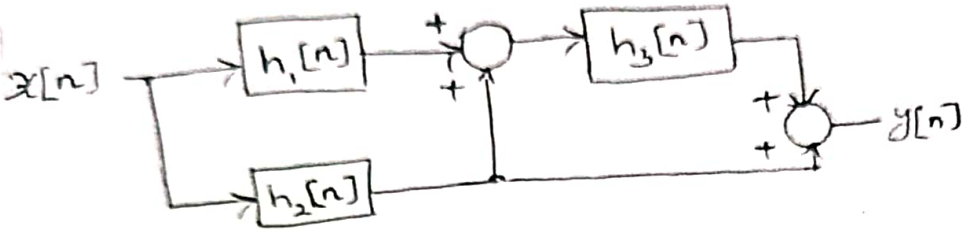
**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**
**MANIPAL INSTITUTE OF TECHNOLOGY, MANIPAL**
**THIRD SEMESTER – MIDTERM TEST**
**SUB: Signals & Systems (ECE-2123)**

**Time:** 90 minutes **Date:** 28/09/2024 **Max Marks:** 30

| Q. No. | Questions | M* | C* | A* | B* |
|---|---|---|---|---|---|
| 1. | Let $x(t)$ be a signal with $x(t) = 0,\ for\ t < 3$. For the signal $g(t) = x(1-t) + x(2-t)$, the value of $t$, for which it is guaranteed to be zero is<br>a. $t > -2$ b. $t < -2$ c. $t > -1$ d. $t < -1$ | 1 | 1 | 1 | 3 |
| 2. | The input $x(t)$ and the output $y(t)$ of a continuous-time system are related as<br>$y(t) = \int_{t-T}^{t} x(u)\, du$<br>The system is<br>(a) Linear and time-variant (b) Linear and time-invariant<br>(b) non-linear and time-variant(d) non-linear and time-invariant | 1 | 2 | 2 | 3 |
| 3. | The spectrum of $x[n] = \frac{\sin(\frac{\pi}{4}n)}{\pi n} * \frac{\sin(\frac{\pi}{2}n)}{\pi n}$, is equal to | 1 | 4 | 4 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | a. $\begin{cases} 1, & for\ \lvert\Omega\rvert \le \pi/4 \\ 0, & for\ \frac{\pi}{4} < \lvert\Omega\rvert \le \pi \end{cases}$<br>b. $\begin{cases} 1, & for\ \lvert\Omega\rvert \le \pi/2 \\ 0, & for\ \frac{\pi}{2} < \lvert\Omega\rvert \le \pi \end{cases}$<br>c. $\begin{cases} 0, & for\ \lvert\Omega\rvert \le \pi/4 \\ 1, & for\ \frac{\pi}{4} < \lvert\Omega\rvert \le \pi \end{cases}$<br>d. $\begin{cases} 0, & for\ \lvert\Omega\rvert \le \pi/2 \\ 1, & for\ \frac{\pi}{2} < \lvert\Omega\rvert \le \pi \end{cases}$ | | | | |
| 4. | The time domain signal for the following frequency coefficients<br>$X(k) = j\,\delta(k-1) - j\delta(k+1) + \delta(k-3) + \delta(k+3)$ with fundamental frequency $\omega_0 = 4\pi$ is<br>e) $2cos(12\pi t) - 2sin(4\pi t)$<br>f) $2cos(12\pi t) + 2jsin(4\pi t)$<br>g) $2sin(12\pi t) - 2cos(4\pi t)$<br>h) $2cos(12\pi t) - 2jsin(\pi t)$ | 1 | 1 | 1 | 3 |
| 5. | The inverse Fourier representation of $X(\Omega) = \frac{1}{5}, \Omega_0 = \frac{2\pi}{10}$ is<br>(a) $x[n] = 2\sum_{l=-\infty}^{\infty} \delta[n-10l]$<br>(b) $x[n] = 2\sum_{l=-\infty}^{\infty} \delta[n-5l]$<br>(c) $x[n] = \sum_{l=-\infty}^{\infty} \delta[n-10l]$<br>(d) $x[n] = \sum_{l=-\infty}^{\infty} \delta[n-5l]$ | 1 | 3 | 3 | 3 |
| 6. | Certain Continuous time LTI system has unit step response $S(t) = (1 - e^{-t})\,u(t)$. Using time domain analysis, determine and sketch the response of the system for the pulse input given by $x(t) = u(t) - u(t-1)$. | 4 | 2 | 2 | 3 |
| 7. | Determine the time domain signal corresponding to the Fourier coefficients $X(k) = cos(k4\pi/11) + 2jsin(k6\pi/11)$. Determine the | 3 | 3 | 3 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | power spectral density, hence using the power spectral density compute the average power of the signal. | | | | |
| 8. | Determine the autocorrelation function and the energy spectral density (ESD) of $x(t) = e^{-2t}u(t)$. Determine the energy of the signal using its ESD. | 3 | 1 | 1 | 3 |
| 9. | Certain discrete time LTI system is described by input-output relation $y[n] = (1/3)\{x[n] + x[n-1] + x[n-2]\}$. Determine its unit step response. | 3 | 4 | 4 | 3 |
| 10. | Find the Fourier representation of $x[n] = n\, e^{j\frac{\pi}{4}} \left(\frac{1}{2}\right)^n u[n-5]$ | 3 | 3 | 3 | 3 |
| 11. | For the given signal $x(t)$ as shown below, sketch the following signals<br>  <br><br>(a) $y_1(t) = x(2t)$<br>(b) $y_2(t) = x(2t+4)$<br>(c) $y_3(t) = x(-\frac{t}{2}+2)$ | 3 | 1 | 1 | 3 |
| 12. | Find the frequency domain representation of the signal shown in the figure below:<br>  <br> | 2 | 1 | 3 | 3 |
| 13. | | 2 | 2 | 2 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | Determine whether the system $y(t) = e^{x(t)}$ is linear, time-invariant, memory, causal, and stable | | | | |
| 14 | Find the overall impulse response for the interconnection of subsystems shown. Given, $h_1[n] = \delta[n]$, $h_2[n] = (1/2)^n u[n-1]$, and $h_3[n] = (-1/2)\ \delta[n-1]$   | 2 | 4 | 4 | 3 |